

36

- Q.3 Define crystal lattice, plane lattice and space lattice.
- Q.4 What is crystal structure?
- Q.5 Define a primitive unit cell. Can a unit cell be primitive?
- Q.6 Define mathematically a lattice. What conditions an ideal lattice must follow?
- Q.7 Define translation vector, unit cell, co-ordination number and packing efficiency.
- Q.8 Write coordination number of simple cubic, body centered cubic and face centered cubic lattices.
- Q.9 What is coordination number of diamond crystal? Also state the number of next nearest neighbour.
- Q.10 What is coordination number of h.c.p structure?
- Q.11 How many lattice points are there in the cubic unit cell of b.c.c structure.
- Q.12 Explain diamond structure. Calculate its packing fraction. Give examples of any two materials having this structure.
- Q.13 Why X-rays are used for crystal structure analysis?
- Q.14 Why γ -rays are not used to study crystal structure?
- Q.15 What is Bragg's law?
- Q.16 What is Bragg's equation?
- Q.17 How does Bragg reflection differ from ordinary reflection?
- Q.18 Explain the term bonding.
- Q.19 What are different kinds of bonding?
- Q.20 What is ionic crystals?
- Q.21 Name various point defects in solids.

PRACTICE PROBLEMS

General Questions

- Q.1 (a) Distinguish between a crystal and an amorphous solid.
(b) Give three main differences between crystalline and amorphous solids.
- Q.2 (a) What is crystal structure? State the relation between crystal structure, lattice and basis.
(b) Define a primitive cell. Distinguish between a primitive unit cell and non-primitive unit cell with the help of diagram. Can a unit cell be primitive?
- Q.3 What is the concept of Miller indices? Derive the formula for the distance between two adjacent planes of a simple cubic lattice.
- Q.4 What is Bravais lattice? Explain different types of Bravais lattices in three dimensions.
- Q.5 Draw the diagrams of the following structures: NaCl and CsCl. Give at least two examples of each structure.
- Q.6 Explain the crystal structure of sodium chloride (NaCl). Draw a sketch of sodium chloride lattice and write down the co-ordinates of the atoms in the unit cell. What is the number of sodium ions in unit cell of NaCl?
- Q.7 Explain the Crystal structure of diamond. In diamond crystal, what is the number of nearest neighbours, the number of atoms per unit cell and packing fraction? Show that it has comparatively loose packing.

- Q.10 Hexagonal closed pack structure is
 (a) simple cubic structure (b) body centered cubic structure
 (c) face centered cubic structure (d) base centered cubic structure
- Q.11 Which one of the following crystals is an example of monoclinic-
 (a) NaCl (b) Mg (c) CaSO_4 (d) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- Q.12 Which one of the following crystals is the example of rhombohedral (or trigonal)-
 (a) CsCl (b) As (c) Fe_3C (d) $\text{K}_2\text{Cr}_2\text{O}_7$
- Q.13 The atomic radius for fcc lattice is
 (a) $\frac{a}{2}$ (b) $\frac{a}{2\sqrt{2}}$ (c) $\frac{\sqrt{3}}{4}a$ (d) $\frac{a}{\sqrt{2}}$
- Q.14 The packing fraction has maximum value for
 (a) fcc structure (b) sc structure
 (c) hpc structure (d) bcc structure
- Q.15 NaCl crystal has
 (a) fcc structure (b) hpc structure
 (c) sc structure (d) bcc structure
- Q.16 Electrons are transferred from one type of atom to the other type in
 (a) covalent crystals (b) ionic crystals
 (c) metallic crystals (d) molecular crystals
- Q.17 van der Waals bond is formed due to
 (a) the sharing of valence electrons
 (b) attraction between positive ion cores and electrons
 (c) electrostatic attraction between oscillating or permanent dipoles
 (d) electrostatic force of attraction between ions
- Q.18 Which one of the followings bonds has high electrical conductivity?
 (a) Ionic bond (b) Covalent bond
 (c) Metallic bond (d) H-bond
- Q.19 The energy required to separate the atoms of a molecule is called
 (a) ionisation energy (b) Coulomb energy
 (c) cohesive energy (d) dissociation energy
- Q.20 If n be the number of atoms in a unit cell of the cubic system, N and M be the Avogadro's number and atomic weight, respectively, and ρ be the density of the element, then the lattice constant a is given by
 (a) $\left[\frac{nM}{N\rho}\right]^{1/3}$ (b) $\left[\frac{nM}{N\rho}\right]^{-1/3}$ (c) $\left[\frac{Mp}{nN}\right]^{1/3}$ (d) $\frac{nM}{N\rho}$

SHORT-ANSWER QUESTIONS

- Q.1 What is a crystal?
- Q.2 Why are most solids crystalline in nature?

10 24

Example 23 Find the ratio of number of Schottky defects to the total number of cation - anion pairs for a binary ionic crystal of NaCl of the average energy required to produce a Schottky defect be 2.02 eV at room temperature. Given Boltzman constant $k = 1.38 \times 10^{-23} \text{ J/K}$.

Solution The number of Schottky defects is given by $n = N \exp\left(\frac{-U}{2kT}\right)$, where N is the total number of cation - anion pairs.

Room temperature $T = 23^\circ\text{C} = 300\text{K}$.

$$\text{Hence } \frac{n}{N} = \exp\left(\frac{-2.02 \times 1.6 \times 10^{-19}}{2 \times 1.38 \times 10^{-23} \times 300}\right) = 1.12 \times 10^{-17}$$

OBJECTIVE TYPE QUESTIONS

- Q.1** Which one of the following Bravais Lattices is not found in cubic crystals?
 (a) Simple cubic (b) Body centered cubic
 (c) Face centered cubic (d) Base centered cubic
- Q.2** In Hexagonal crystal, the angles between the axes are
 (a) $\alpha = \beta = \gamma = 90^\circ$ (b) $\alpha \neq \beta \neq \gamma \neq 90^\circ$
 (c) $\alpha = \beta = 90^\circ, \gamma = 120^\circ$ (d) $\alpha = \gamma = 90^\circ \neq \beta$
- Q.3** The coordination number of b.c.c lattice is
 (a) 6 (b) 8
 (c) 12 (d) 3
- Q.4** The number of atoms present in the f.c.c lattice is
 (a) 1 (b) 2 (c) 3 (d) 4
- Q.5** The packing fraction of diamond crystal structure is
 (a) 74% (b) 68% (c) 52% (d) 34%
- Q.6** The number of ions pair per unit cell present in NaCl is
 (a) 5 (b) 2 (c) 4 (d) 1
- Q.7** X-rays can be deflected by
 (a) magnetic fields (b) electric fields
 (c) large size obstacles (d) none of these
- Q.8** The Miller indices of a set of parallel planes which make equal intercepts on the three axes are
 (a) (1 2 1) (b) (1 1 1) (c) (1 0 0) (d) (1 0 1)
- Q.9** In a simple cubic lattice the ratio $d_{100}:d_{110}:d_{111}$ is
 (a) $\sqrt{6}:\sqrt{3}:\sqrt{2}$ (b) $\sqrt{3}:\sqrt{6}:1$ (c) $6:3:\sqrt{2}$ (d) $6:3:1$

$$a^3 = \frac{nM}{N\rho}$$

$$\text{and } d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \text{ for cubic crystal.}$$

$$\therefore a^3 = \frac{nM}{N\rho} = \frac{4 \times 74.5}{6.023 \times 10^{26} \times 1.98 \times 10^3} = 24.99 \times 10^{-29}$$

$$a = [249.9 \times 10^{-30}]^{1/3} = 6.30 \times 10^{-10} = 6.3 \text{ \AA}$$

$$d_{100} = \frac{a}{\sqrt{1^2 + 0^2 + 0}} = \frac{a}{1} = \frac{6.3 \text{ \AA}}{1} = 6.3 \text{ \AA}$$

$$d_{110} = \frac{a}{\sqrt{1^2 + 1^2 + 0}} = \frac{a}{\sqrt{2}} = 4.38 \times 10^{-10} \text{ m} = 4.38 \text{ \AA}$$

$$d_{111} = \frac{a}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{a}{\sqrt{3}} = \frac{6.3 \times 10^{-10}}{\sqrt{3}} = 3.64 \times 10^{-10} \text{ m} = 3.64 \text{ \AA}$$

Example 21 Determine the potential energy of K^+ and Cl^- ion when they are separated by a distance of 0.15 nm.

Solution Given separation distance $r_0 = 0.15 \text{ nm} = 0.15 \times 10^{-9} \text{ m}$.

$$\text{The potential energy of the ions } (V) = \frac{-e^2}{4\pi\epsilon_0 r_0} \quad J = \frac{-e}{4\pi\epsilon_0 r_0} \text{ eV}$$

$$\begin{aligned} \therefore V &= -\frac{1.6 \times 10^{-19}}{4 \times 3.14 \times 8.85 \times 10^{-12} \times 0.15 \times 10^{-9}} \\ &= -0.0959 \times 10^2 \text{ eV} \\ &= -0.096 \times 10^2 \text{ eV} = -9.6 \text{ eV} \end{aligned}$$

Example 22 From the following data determine the cohesive energy of NaCl. The equilibrium separation $r_0 = 0.32 \text{ nm}$, $\alpha = 1.748$, $n = 9$, ionisation energy = 4 eV and electron affinity = -2.16 eV.

Solution Given $r_0 = 0.32 \text{ nm} = 0.32 \times 10^{-9} \text{ m}$, $\alpha = 1.748$, $n = 9$, Ionisation energy = 4 eV and electron affinity = -2.16 eV.

$$\begin{aligned} V(r_0) &= \frac{-\alpha e^2}{4\pi\epsilon_0 r_0} \left(1 - \frac{1}{n}\right) \text{ Joule or } V(r_0) = \frac{-\alpha e}{4\pi\epsilon_0 r_0} \left(1 - \frac{1}{n}\right) \text{ electron volt} \\ &= \frac{-1.748 \times (1.6 \times 10^{-19})^2}{4 \times 3.14 \times 8.85 \times 10^{-12} \times 0.32 \times 10^{-9}} \left(1 - \frac{1}{9}\right) \\ &= 0.0698 \text{ eV} \end{aligned}$$

Thus, $2 \times 3.84 \times 10^{-10} \times \sin 30^\circ = \lambda$

or $\lambda = 2 \times 3.84 \times 10^{-10} \times \frac{1}{2} = 3.84 \text{ \AA}$

$\therefore \lambda = \frac{h}{mv}$ [according to deBroglie relation]

$\therefore \lambda = \frac{h}{m\lambda} = \frac{6.62 \times 10^{-34}}{1.67 \times 10^{-27} \times 3.84 \times 10^{-10}}$

$\lambda = 1.03 \times 10^{-8} \text{ m}$

Example 19 Electrons accelerated from the state of rest by 120V are reflected from an fcc crystal. The reflection maximum is observed at 22° . Determine the lattice parameter if the Bragg reflection occurs from the (1 1 1) plane.

Solution Given $V = 120\text{V}$, $\theta = 22^\circ$ and $n = 1$.

Formula used are $\lambda = \frac{h}{(2meV)^{1/2}}$ $\left[\because \frac{1}{2}mv^2 = eV \right]$

and $2d \sin \theta = n\lambda$.

$\therefore \lambda = \frac{h}{\sqrt{2meV}}$

$$= \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 120}} = 1.12 \times 10^{-10} \text{ m}$$

so, $d_{111} = \frac{n\lambda}{2\sin\theta} = \frac{1 \times 1.12}{2 \times \sin 22^\circ} = 1.4949 \text{ \AA}$

and $d_{111} = \frac{a}{\sqrt{3}}$

$\therefore a = d_{111} \times \sqrt{3} = \sqrt{3} \times 1.4949 \text{ \AA} = 2.589 \text{ \AA}$

Example 20 A monochromatic beam of X-rays of wavelength $1.24 \text{ \AA}$ is reflected by cubic crystal of KCl. Determine the inter planar distances for (1 0 0), (1 1 0) and (1 1 1) planes. Given density of KCl = 1980 kg/m^3 and molecular weight $M = 74.5$. Avogadro's number $N = 6.023 \times 10^{26} / \text{kg mole}$.

Solution Given $M = 74.5$, $\rho = 1980 \text{ kg/m}^3$ and $N = 6.023 \times 10^{26} / \text{kg mole}$.

Formulas used are

$$\therefore d = \left(\frac{n\lambda}{2 \sin \theta} \right) = \frac{1 \times 1.5 \times 10^{-10}}{2 \sin 60^\circ} = \frac{1.5 \times 10^{-10} \times 2}{2 \times \sqrt{3}} = 0.87 \text{ \AA}$$

Example 16 X-ray of wavelength $1.4 \text{ \AA}$ is found to be Bragg reflected from the $(1\ 1\ 1)$ plane of an fcc structure. If the lattice parameter of the crystal is $5 \text{ \AA}$. Find the angle at which the X-ray is incident on the $(1\ 1\ 1)$ plane of the crystal.

Solution Given $\lambda = 1.4 \text{ \AA}$, lattice parameter of fcc structure (a) = $5 \text{ \AA}$ and the plane of the fcc structure = $(1\ 1\ 1)$.

Interplanar spacing $d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$

and $2d \sin \theta = n\lambda$

$$\therefore d_{111} = \frac{a}{\sqrt{3}} = \frac{5 \times 10^{-10}}{\sqrt{3}} = 2.887 \times 10^{-10} \text{ m}$$

$$\text{and so, } \theta_{111} = \sin^{-1} \left(\frac{n\lambda}{2d_{111}} \right) = \sin^{-1} \left(\frac{1 \times (1.4 \times 10^{-10})}{2 \times 2.887 \times 10^{-10}} \right) = 14^\circ$$

Example 17 Calculate the glancing angle on the cube face $(1\ 0\ 0)$ of a rock salt crystal ($a = 2.814 \text{ \AA}$) corresponding to second order reflection of X-rays of wavelength $0.710 \text{ \AA}$.

Solution Given $d = a = 2.814 \text{ \AA} = 2.814 \times 10^{-10} \text{ m}$ for cube face $(1\ 0\ 0)$, $n = 2$ for second order diffraction and $\lambda = 0.710 \times 10^{-10} \text{ m}$.

$$2d \sin \theta = n\lambda$$

$$\begin{aligned} \theta &= \sin^{-1} \left[\frac{n\lambda}{2d} \right] \\ &= \sin^{-1} \left[\frac{2 \times 0.710 \times 10^{-10}}{2 \times 2.814 \times 10^{-10}} \right] \\ &= 14.6^\circ \end{aligned}$$

Example 18 From the following data calculate the wavelength of neutron beam and its speed. Spacing between successive planes is $3.84 \text{ \AA}$, glancing angle 30° and the order of Bragg reflection = 1.

Solution Given $d = 3.84 \text{ \AA} = 3.84 \times 10^{-10} \text{ m}$, $\theta = 30^\circ$ and $n = 1$.

Formula used are $2d \sin \theta = n\lambda$ and $\lambda = \frac{h}{mv}$.

lattice

$$\begin{aligned}
 a &= \left[\frac{4 \times 60.2}{6250 \times 6.02 \times 10^{26}} \right]^{1/3} \\
 &= \left[\frac{240.8}{37265 \times 10^{26}} \right]^{1/3} \\
 &= 4 \text{ \AA}
 \end{aligned}$$

Example 13 In NaCl crystal, the spacing between the successive (1 0 0) plane is 2.82 Å. X-ray incident on the surface of the crystal is found to give rise to first order Bragg reflection at glancing angle 8.8°. Calculate the wavelength of X-ray.

Solution Given $d = 2.82 \text{ \AA}$, $\theta = 8.8^\circ$ and $n = 1$.

Formula used is $2d \sin \theta = n\lambda$.

$$\begin{aligned}
 2 \times 2.82 \times 10^{-10} \times \sin 8.8^\circ &= \lambda \\
 \Rightarrow \lambda &= 5.64 \times 10^{-10} \times 0.153 \\
 &= 0.863 \text{ \AA}
 \end{aligned}$$

Example 14 The first-order diffraction is found to occur at a glancing angle of 9° . Calculate the wavelength of X-ray and the glancing angle for second order diffraction if the spacing between the adjacent plane is 2.51 Å.

Solution Given $n = 1$, $\theta = 9^\circ$ and $d = 2.51 \text{ \AA} = 2.51 \times 10^{-10} \text{ m}$.

Formula used, is $2d \sin \theta = n\lambda$.

Therefore, $\lambda = 2 \times (2.51 \times 10^{-10}) \times \sin 9^\circ = 0.7853 \text{ \AA}$.

For $n = 2$

$$\begin{aligned}
 \theta &= \sin^{-1} \left(\frac{2\lambda}{2d} \right) = \sin^{-1} \left(\frac{\lambda}{d} \right) \\
 &= \sin^{-1} \left(\frac{0.7853 \times 10^{-10}}{2.51 \times 10^{-10}} \right) \\
 &= 18.2^\circ
 \end{aligned}$$

Example 15 X-rays of wavelength 1.5 Å make a glancing angle 60° in the first order when diffracted from NaCl crystal. Find the lattice constant of NaCl.

Solution Given $\lambda = 1.5 \text{ \AA}$, $\theta = 60^\circ$ and $n = 1$.

Formula used is $2d \sin \theta = n\lambda$.

Example 9 Calculate the distance between two atoms of basis of the diamond structure if the lattice constant of the structure is $5\text{\AA}$.

Solution Given lattice constant $a = 5\text{\AA}$.

The distance between two atoms is equivalent to the nearest neighbor distance.

For diamond structure, nearest neighbour distance $= \frac{\sqrt{3}}{4}a$

$$\therefore \text{Distance between two atoms} = \frac{\sqrt{3}}{4} \times 5\text{\AA} = \frac{1.732 \times 5}{4} = 2.17\text{\AA}$$

Example 10 What is the number of atoms in the primitive cell of diamond. Calculate the length of primitive translation vector if the cube edge $a = 3.56\text{\AA}$.

Solution Diamond is a fcc lattice with two carbon atoms in a primitive cell. So, the number of atoms is 8.

Given the cube edge $a = 3.56\text{\AA}$.

$$\therefore \text{Primitive translation vector} = \frac{a}{\sqrt{2}} = \frac{3.56}{1.41} = 2.52\text{\AA}$$

Example 11 Determine the number of atoms per unit cell of lead which has an fcc structure. Atomic weight of Pb = 207.2, density of Pb = $11.36 \times 10^3 \text{ kg m}^{-3}$, and $a = 3.2\text{\AA}$ and Avogadro's number = $6.023 \times 10^{26} / \text{kg mole}$.

Solution Given atomic weight of Pb (M) = 207.2, density of Pb = $11.36 \times 10^3 \text{ kg m}^{-3}$ and $a = 3.2\text{\AA}$

Avogadro's number $N = 6.023 \times 10^{26} / \text{kg mole}$.

$$\text{Number of atoms } n = \frac{a^3 \rho N}{M}$$

$$= \frac{3.2 \times 10^{-10} \times 11.36 \times 10^3 \times 6.023 \times 10^{26}}{207.2}$$

$$= 1.082 \approx 1$$

Example 12 Calculate the lattice constant 'a' of a substance having fcc lattice, molecular weight 60.2 and density 6250 kg/m^3 . ($N = 6.02 \times 10^{26} / \text{kg mole}$)

Solution Given molecular weight $M = 60.2$, density $\rho = 6250 \text{ kg/m}^3$, and $N = 6.02 \times 10^{26} / \text{kg mole}$.

For fcc lattice $n = 4$

$$\text{Lattice constant } a = \left(\frac{4M}{N\rho} \right)^{1/3}$$

Example 7 Calculate the interplanar spacing for (2 3 1) plane of an fcc structure whose atomic radius is 0.175 nm.

Solution Given plane = (2 3 1) and atomic radius (r) = 0.175 nm.

Atomic radius (r) of fcc structure

$$= \frac{\sqrt{2}}{4} a$$

and interplanar spacing $d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$

$$\begin{aligned} \therefore r = \frac{\sqrt{2}a}{4} \Rightarrow a &= \frac{4r}{\sqrt{2}} = \frac{4 \times 0.175 \times 10^{-9}}{\sqrt{2}} \\ \therefore d_{231} &= \frac{a}{\sqrt{h^2 + k^2 + l^2}} = \frac{4 \times 0.175 \times 10^{-9}}{\sqrt{2 \times 2^2 + 3^2 + 1}} \\ &= \frac{4 \times 0.175 \times 10^{-9}}{\sqrt{2} \times \sqrt{14}} = 0.132 \times 10^{-9} \text{ m} \\ d_{231} &= 0.132 \text{ nm} \end{aligned}$$

Example 8 In a simple cubic crystal (i) find the ratio of intercepts of three axes by (1 2 3) plane and (ii) find the ratio of spacings of (1 1 0) and (1 1 1) planes.

Solution (i) Given ($h k l$) of the plane as (1 2 3). Intercepts on the axes of a simple cubic crystal are given as $a/h, a/k, a/l$.

$\therefore$ The ratio of intercepts are

$$\frac{a}{1} : \frac{a}{2} : \frac{a}{3} = 1 : \frac{1}{2} : \frac{1}{3}$$

(ii) The spacings d of plane (h, k, l) in a simple cubic crystal of side a

$$= \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$\therefore \text{ For plane (1 1 0), } d_{110} = \frac{a}{\sqrt{1^2 + 1^2 + 0}} = \frac{a}{\sqrt{2}}$$

$$\text{ and for plane (1 1 1), } d_{111} = \frac{a}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{a}{\sqrt{3}}$$

Therefore the ratio of spacing between these two plane is $d_{110} / d_{111} = \sqrt{3/2} = 1.225$

13.26 Engineering Physics

For plane (1 0 0), $d_{100} = \frac{a}{\sqrt{1^2 + 0^2 + 0^2}} = a$

and For plane (1 1 1), $d_{111} = \frac{a}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{a}{\sqrt{3}}$

Example 5 Deduce the Miller indices of a set of parallel planes which make intercepts in the ratio of $a:2b$ or the x and y axis and are parallel to z -axis, $\vec{a}, \vec{b}, \vec{c}$ being primitive vectors of lattice. Also calculate the interplanar distance d of the plane taking the lattice to be cubic with $a = b = c = 5\text{\AA}$.

Solution The parallel planes are parallel to z axis. It means that their intercepts on the z -axis are infinite. Thus, the intercepts are $a, 2b$ and ∞ . And, lattice constant $a = b = c = 5\text{\AA}$.

Then,

$$\begin{aligned} a:2b:\infty c &= \frac{a}{h}:\frac{b}{k}:\frac{c}{l} \\ \text{or } \frac{1}{h}:\frac{1}{k}:\frac{1}{l} &= 1:2:\infty \\ \text{or } h:k:l &= 1:\frac{1}{2}:\frac{1}{\infty} = 2:1:0 \end{aligned}$$

Therefore, the Miller indices are (210).

$\therefore$ Interplanar distance

$$\begin{aligned} d &= \frac{a}{\sqrt{h^2 + k^2 + l^2}} = \frac{5 \times 10^{-10}}{\sqrt{2^2 + 1^2 + 0^2}} \\ &= \frac{5 \times 10^{-10}}{\sqrt{5}} = \sqrt{5} \text{\AA} \end{aligned}$$

Example 6 Determine the Miller indices of plane parallel to the z axis and cut intercepts of 2 and $2/3$ along x and y axes, respectively.

Solution Intercepts are $2a, \frac{2b}{3}, \infty$.

$$\begin{aligned} 2a:\frac{2b}{3}:\infty c &= \frac{a}{h}:\frac{b}{k}:\frac{c}{l} \\ \text{or } \frac{1}{h}:\frac{1}{k}:\frac{1}{l} &= 2:\frac{2}{3}:\infty \\ \text{or } h:k:l &= \frac{1}{2}:\frac{3}{2}:\frac{1}{\infty} = 1:3:0 \end{aligned}$$

Therefore, the Miller indices are (1 3 0).

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26

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Example 3 Deduce the Miller indices for planes in each of the following sets which intercept $\vec{a}$, $\vec{b}$ and $\vec{c}$ at (i) $3a$, $3b$, $2c$ (ii) a , $2b$, ∞ (iii) a , $b/2$, c

aCl at ordinary

Solution (i) Intercepts are $3a$, $3b$, $2c$.

f cation-anion Then,

$$\begin{aligned} 3a : 3b : 2c &= \frac{a}{h} : \frac{b}{k} : \frac{c}{l} \\ \text{or } 3 : 3 : 2 &= \frac{1}{h} : \frac{1}{k} : \frac{1}{l} \\ \text{or } h : k : l &= \frac{1}{3} : \frac{1}{3} : \frac{1}{2} \\ \text{or } h : k : l &= 2 : 2 : 3 \end{aligned}$$

s given by

f interstitial

Therefore, the Miller indices are $(2\ 2\ 3)$.

(ii) Intercepts are a , $2b$, ∞ .

termine the Then,

$$\begin{aligned} a : 2b : \infty &= \frac{a}{h} : \frac{b}{k} : \frac{c}{l} \\ \text{or } \frac{1}{h} : \frac{1}{k} : \frac{1}{l} &= 1 : 2 : \infty \\ \text{or } h : k : l &= 1 : \frac{1}{2} : \frac{1}{\infty} = 2 : 1 : 0 \end{aligned}$$

Therefore, the Miller indices are $(2\ 1\ 0)$.

(iii) Intercepts are a , $b/2$, c .

Then,

$$\begin{aligned} a : \frac{b}{2} : c &= \frac{a}{h} : \frac{b}{k} : \frac{c}{l} \\ \text{or } 1 : \frac{1}{2} : 1 &= \frac{1}{h} : \frac{1}{k} : \frac{1}{l} \\ \text{or } h : k : l &= 2 : 1 : 2 \end{aligned}$$

Therefore, the Miller indices are $(2\ 1\ 2)$.

Example 4 Calculate the spacing between $(1\ 0\ 0)$ and $(1\ 1\ 1)$ planes of a cubic system of lattice parameter a .

Solution Spacing between the planes of a cubic system of lattice parameter a .

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

25

defect. Point defect occurs because of the absence of a matrix atom or the presence of an impurity atom at the matrix atom in the wrong place.

- (16) The number of Schottky defects in binary ionic crystals like MgO and NaCl at ordinary temperature is given by $n = N \exp\left(-\frac{U}{2kT}\right)$, where N is the total number of cation-anion pairs and U is the average energy required to produce the Schottky defects.
- (17) The number of Frenkel defects in crystals at ordinary temperature is given by $n = (NN_i)^{1/2} \exp\left(-\frac{E_f}{2kT}\right)$, where N is the number of ions, N_i is the number of interstitial sites and E_f is the energy required to produce the vacancy.

SOLVED EXAMPLES

Example 1 A plane cuts intercepts $2a$, $3b$ and c along the crystallographic axes in a crystal. Determine Miller indices of plane.

Solution Intercepts are $2a$, $3b$ and c .

Then from the law of rational indices, we have

$$2a:3b:c = \frac{a}{h}:\frac{b}{k}:\frac{c}{l}$$

$$\text{or } \frac{1}{h}:\frac{1}{k}:\frac{1}{l} = 2:3:1$$

$$\text{or } h:k:l = \frac{1}{2}:\frac{1}{3}:1 = 3:2:6$$

Therefore, the Miller indices of the plane are $(3\ 2\ 6)$.

Example 2 In a triclinic crystal, a lattice plane makes intercepts at a length a , $2b$ and $-3c/2$. Find the Miller indices of the plane.

Solution Intercepts are a , $2b$ and $-3c/2$.

$$\therefore a:2b:-\frac{3c}{2} = \frac{a}{h}:\frac{b}{k}:\frac{c}{l}$$

$$\text{or } \frac{1}{h}:\frac{1}{k}:\frac{1}{l} = 1:2:-\frac{3}{2}$$

$$\text{or } h:k:l = 1:\frac{1}{2}:-\frac{2}{3} = 6:3:-4$$

Therefore, the Miller indices of the given plane are $(6\ 3\ \bar{4})$.

- 24
- (6) A crystal lattice is made of large number of parallel equidistant planes known as lattice planes.
- (7) The integers which determine the orientation of a crystal plane in relation to the three crystallographic axes are called Miller indices. In order to find the Miller indices, the reciprocals of the intercepts of the plane on the axes in terms of lattice constants are reduced to the smallest integers in ratio. Miller indices are also called crystal indices.
- (8) In a crystal, every atom is surrounded by the other atoms. The number of nearest neighbours to the given atom in the crystal lattice is known as coordination number.
- (9) For a set of planes ($h k l$) in a unit cell, the distance between adjacent planes or interplanar spacing between parallel planes is given by $d = \frac{1}{\sqrt{\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}}$. Here a , b and c are the fundamental translational vectors along the three axes.
- (10) The distance between the centers of two neighbouring atoms is called the nearest neighbour distance. For a closely packed crystal, this distance is $2r$ for an atom of radius r . The distance r is called atomic radius, which is generally represented in terms of edge of cube a for certain unit cell structures.
- (11) The ratio of volume of atoms occupying the unit cell to the volume of the unit cell relating to that substance is called atomic packing fraction. It is also known as relative packing density. It is denoted as f . The atomic packing fraction f for the simple cubic structure is 52%. For a body centered cubic (bcc) structure $f = 68\%$ and for a face centered cubic (fcc) and hexagonal closed packed structures $f = 74\%$. The atomic packing fraction f for the diamond structure is 34%.
- (12) The interatomic forces between atoms of the solids are electrostatic in nature which can either be attractive or repulsive. It is obtained that at a particular distance, the attractive and repulsive forces attain equal values and therefore no force acts between the atoms.
- (13) The constituent particles of crystals have different types of charge distribution which provides different types of binding forces. The binding forces in most cases are electrostatic in nature but the distribution of electrons in various atoms are qualitatively different in different crystals. These binding forces are of different types, for example, ionic bond, covalent bond, metallic bond, molecular bond (or van der Waals bonds) and hydrogen bonds. Accordingly, the crystals are referred to as the ionic crystal, covalent crystal, metallic crystal, molecular crystal and hydrogen bonded crystal.
- (14) Since the X-rays can penetrate solids and their wavelength ($1 \text{ \AA}$) is of the order of interplanar spacing, these rays can get strongly diffracted from different crystal planes. An analysis of the diffracted X-rays can provide the information about the structure of the crystal. The standard methods of X-ray diffraction include Bragg's X-ray spectrometer, Laue method, rotating crystal method and powder method.
- (15) Point defect is a discontinuity in a crystal lattice. It consists of either a missing atom or an ion that creates a vacancy in the lattice (often known as Schottky defect). If an extra atom or ion exists between two normal lattice points, it is said to create an interstitial position and if the missing atom or ion shifts to an interstitial position, then the vacancy is called Frenkel

23

13.22 Engineering Physics

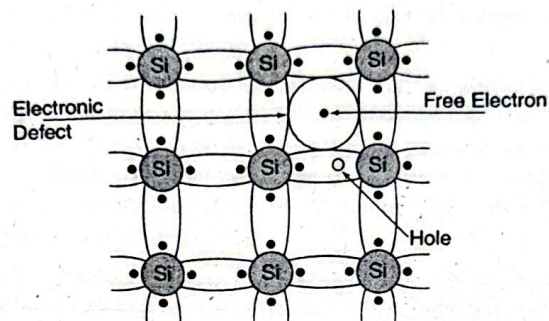


Fig. 13.33

13.17 SUMMARY

The topics covered in this chapter are summarised below.

- (1) A unique arrangement of atoms is called a crystal structure. It consists of a set of atoms which are identical in composition, arrangement and orientation called basis and a lattice. Bases are located upon the points of a lattice, which is an array of points repeating periodically in three dimensions. A crystal structure and symmetry play an important role in determining many of its properties such as electronic band structure and optical properties.
- (2) The solids are broadly classified into two groups, namely crystalline solids and amorphous solids. Crystalline solids are arranged in fixed geometric patterns or lattices. Ice, methanol and sodium chloride are a few examples of crystalline solids. A rigid material whose structure lacks crystalline periodicity is called an amorphous solid. It means the pattern of its constituent atoms or molecules does not repeat periodically in three dimensions in amorphous solids.
- (3) A crystal structure can be obtained by translation of a unit cell in three dimensions. A unit cell is a smallest pattern of a space lattice, which can generate the complete crystal by repeating its own dimensions in various directions in amorphous solids.
- (4) Based on the shape of the unit cells all crystals are classified into seven crystal systems. These are cubic system, trigonal system, tetragonal system, hexagonal system, orthorhombic system, monoclinic system and triclinic system. In a cubic system, there are three types of lattices called simple cubic (sc) system, body centered cubic (bcc) system and face centered cubic (fcc) system.
- (5) With reference to a lattice point as an origin in a plane lattice, any other point can be obtained by repeatedly translating the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$, which are the primitives along X, Y and Z axes, respectively. The position vector of this lattice point, i.e. translation vector, can be represented as

$$\vec{T} = n_1\vec{a} + n_2\vec{b} + n_3\vec{c},$$

... the number of lattice points along the

Differentiating Eq. (vi) w.r.t. n , we get

$$\left[\frac{\partial F}{\partial n} \right]_T = E_i - kT \ln \frac{(N-n)(N_i-n)}{n^2} \quad (\text{vii})$$

Free energy remains constant, when the equilibrium position is attained at a given temperature T . It means

$$\left[\frac{\partial F}{\partial n} \right]_T = 0$$

$$\therefore E_i - kT \ln \frac{(N-n)(N_i-n)}{n^2} = 0.$$

$$\text{or} \quad \ln \frac{(N-n)(N_i-n)}{n^2} = \frac{E_i}{kT}.$$

For smaller number of Frenkel defects, i.e. when $N \gg n$ and $N_i \gg n$ the above relation reads

$$\ln \frac{NN_i}{n^2} = \frac{E_i}{kT}$$

$$\text{or} \quad 2 \ln n = \ln(NN_i) - \frac{E_i}{kT}$$

$$\text{or} \quad \ln n = \frac{1}{2} \ln(NN_i) - \frac{E_i}{2kT}$$

$$\text{or} \quad n = (NN_i)^{1/2} e^{-E_i/2kT}$$

From the above relation the concentration of Frenkel defects at a temperature T can be calculated.

13.16.2 Interstitial

As defined earlier, in interstitial defect an atom or an ion moves from its proper position to a place between regular lattice sites, as shown in Fig. 13.31. The interstitial is either due to a normal atom of the crystal or of a foreign atom.

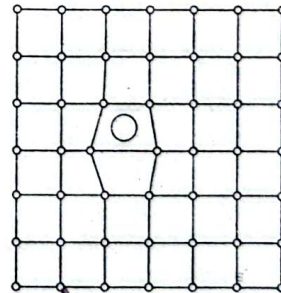


Fig. 13.31

13.16.3 Compositional Defect

The compositional defect occurs because of the replacement of a host atom by a foreign atom. The foreign atom remains at the regular lattice site, as shown in Fig 13.32.

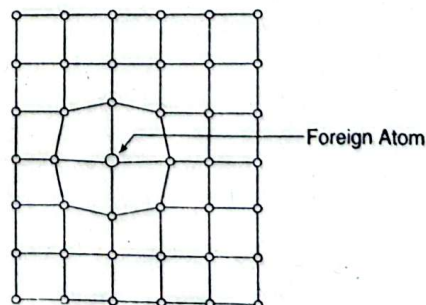


Fig. 13.32

13.16.4 Electronic Defect

At absolute zero in a purely covalent crystal (e.g. Si), the electrons are tightly bound to the core and all are said to be in the valence band. Above absolute zero, some of the electrons are likely to occupy higher energy state depending upon the temperature. So in the crystal of pure silicon, some of the electrons from the covalent bonds get thermally released and become free to move, as shown in Fig. 13.33. This way the deficiency of electron creates a hole. Then the electrons and holes give rise to electronic imperfections.

Q.4 Calculate the glancing angle at which X-rays with $\lambda = 1.549 \text{ \AA}$ will be reflected in first and second orders from a crystal with interplanar distance $4.225 \text{ \AA}$

[Ans: $\theta_1 = 10^\circ 31'$ and $\theta_2 = 21^\circ 21'$]

Q.5 Using 2.02 as the value of lattice constant, calculate the wavelength of X-ray in second order, if angle of diffraction $\theta = 26^\circ$.

[Ans: $1.24 \text{ \AA}$]

Q.6 A crystal is mounted on an X-ray spectrometer. X-rays are incident at the glancing angle for three reflections are $5^\circ 28'$, $12^\circ 1'$ and $18^\circ 12'$. Show that these are successive orders of reflections from the same crystal plane. Also find the spacing. [Given λ for X-rays used as $0.586 \text{ \AA}$].

[Ans: $2.817 \text{ \AA}$, $2.817 \text{ \AA}$ and $2.817 \text{ \AA}$]

Q.7 A certain crystal reflects monochromatic X-rays strongly when Bragg glancing angle (first order) is 15° . What are the glancing angles for second and third order spectrum.

[Ans: 31.17° , 50.93°]

and $\vec{c}$, at (i)

and $(2,1,0)$

and (iii) (110)

(iii) $a/\sqrt{2}$

from the origin

and $a/\sqrt{29}$

13.20 Engineering Physics

This will give $\left. \frac{\partial F}{\partial n} \right|_T = 0$ or $U = 2kT \ln \left[\frac{(N-n)}{n} \right]$
 or $\frac{N-n}{n} = \exp \left(\frac{U}{2kT} \right)$

For smaller number of Schottky defects, i.e. when $n \ll N$, we have $N - n \approx N$.

This gives, $n = N \exp \left(-\frac{U}{2kT} \right)$

The above expression gives the number of Schottky defects in binary ionic crystals like MgO and NaCl at ordinary temperature. It is clear from the expression that the fraction n/N of Schottky defects increases exponentially with the temperature.

Evaluation of Concentration of Frenkel Defects

In order to calculate the concentration of Frenkel defects, we consider a pure crystal that consists of positively and negatively charged ions in equal proportion. Further, we consider that it contains total N number of ions and n Frenkel defects. The number of different ways in which n Frenkel defects will be produced is given as

$$W = \frac{N!}{(N-n)!n!} \cdot \frac{N_i!}{(N_i-n)!n!} \quad (i)$$

where N_i is the number of interstitial sites. The Helmholtz free energy F of the crystal is given by the relation

$$F = U - ST \quad (ii)$$

where S is the increase in entropy and T is the temperature.

If E_i be the energy required to produced a vacancy, then U can be expressed as

$$U = nE_i \quad (iii)$$

and the associated increase in entropy is given by the Boltzmann relation

$$S = k \ln W = k \ln \left[\frac{N!}{(N-n)!n!} \cdot \frac{N_i!}{(N_i-n)!n!} \right] \quad (iv)$$

With the help of Eqs. (iii) and (iv), Eq. (ii) becomes

$$F = nE_i - kT \ln \left[\frac{N!}{(N-n)!n!} \cdot \frac{N_i!}{(N_i-n)!n!} \right] \quad (v)$$

Sterling's approximation $[\ln y! = y \ln y - y]$ yields

$$\begin{aligned} \ln \left[\frac{N!}{(N-n)!n!} \cdot \frac{N_i!}{(N_i-n)!n!} \right] &= N \ln N + N_i \ln N_i - \{(N-n) \ln(N-n) \\ &\quad + (N_i - n) \ln(N_i - n) + 2n \ln n\} \\ &= nE_i - kT[N \ln N + N_i \ln N_i - (N-n) \ln(N-n) \\ &\quad - (N_i - n) \ln(N_i - n) - 2n \ln n] \end{aligned} \quad (vi)$$

Differentiat

Free energy

$$\left[\frac{\partial F}{\partial n} \right]_T = 0$$

For smaller

From the ab
T can be cal

13.16.2

As defined
proper posit
The interstit

13.16.3

The compos
a host atom
regular lattic

13.16.4

At absolute z
are tightly b
band. Above
occupy high
the crystal of
bonds get the
in Fig. 13.33
Then the elec

ies causes to missing of
ion from the structure.
sites is called a *Schottky*
f a cation goes into an
en the interstitially pair
defect (Fig. 13.30).

Concentration of ts

Schottky defect, the
more than compensated
order in the lattice. Due
the disorder increases

cement in entropy. A certain number of vacancies, called the concentration of Schottky
resent in thermal equilibrium which can be calculated as follows.

n is the number of Schottky defects produced in removing na^+ cations and na^- anions.
 N is the total number of cation-anion pairs. If U be the average energy, which is required
y defect, then nU would be the increase in energy associated with the generation of n
er of different ways in which a cation or an anion can be removed is given by $\frac{N!}{(N-n)!n!}$.

ion and n anions, the total number of different ways in which n Schottky defects can be

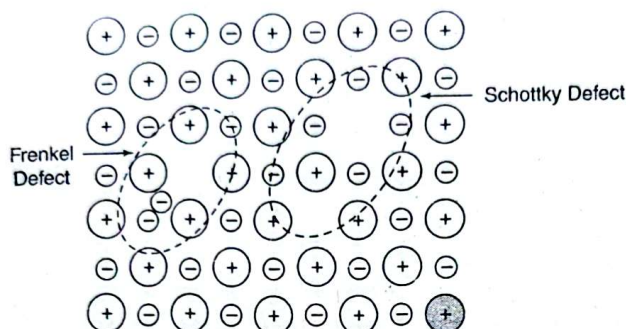


Fig. 13.30

$$= \left[\frac{N!}{(N-n)!n!} \right] \left[\frac{N!}{(N-n)!n!} \right]$$

opy is given by

$$= k \ln W$$

$$= k \ln \left[\frac{N!}{(N-n)!n!} \right]^2$$

ann constant. This increase in entropy produces a change in the Helmholtz free energy
ined as

$$= \text{Increase in energy} - \text{Temperature} \times \text{Increase in entropy}$$

$$= nU - 2kT \ln \left[\frac{N!}{(N-n)!n!} \right]$$

approximation $\ln y! = y \ln y - y$, we get the following expression for the change in
gy F

$$= nU - 2kT [N \ln N - (N-n) \ln (N-n) - n \ln n]$$

e a constant at equilibrium.

13.18 Engineering Physics

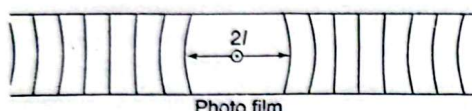


Fig. 13.27

possible diffraction planes will be available for the Bragg diffraction ($2d \sin \theta = n\lambda$) to take place.

All these diffracted rays will lie on a conical surface having its apex at P and semivertical angle 2θ . The diffracted X-ray is recorded by the photographic film placed around the crystal and we get the arc of the circle on the photographic film, as shown in Fig. 13.27.

If $2l$ is the distance of the two arcs of the same circle on the photographic film and R is the radius of cylindrical camera (Fig. 13.28), then

$$2\theta = \frac{l}{R} \quad \text{or} \quad \theta = \frac{l}{2R}$$

This value of θ will give the spacing between the planes with the help of Bragg's relation,

$$2d \sin \theta = n\lambda$$

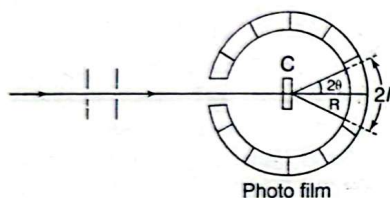


Fig. 13.28

y differentiating

$$2\Delta d \sin \theta + 2d \cos \theta \cdot \Delta \theta = 0$$

$$\text{or} \quad \frac{\Delta \theta}{\Delta d} = -\frac{\tan \theta}{d}$$

θ tends to 90° then $2\theta = 180^\circ$. For this angle, X-rays get reflected back along their initial path and such reflected beams cannot be recorded. For $\theta = 90^\circ$, $(\Delta \theta / \Delta d)$ becomes very large so that small variations in produces large variation in θ .

13.16 POINT DEFECTS IN SOLIDS

Crystal defects: ① Point defects ② line defects ③ surface defects

Point defect is a discontinuity in a crystal lattice. It consists of either a missing atom or an ion that creates a vacancy in the lattice (often known as Schottky defect). If an extra atom or ion exists between two normal lattice points it is said to create an interstitial position and if the missing atom or ion shifts to an interstitial position, then the vacancy is called Frenkel defect. Point defect occurs because of the absence of a matrix atom or the presence of an impurity atom or the matrix atom at the wrong place. The most common type point defects in a pure crystal are given below.

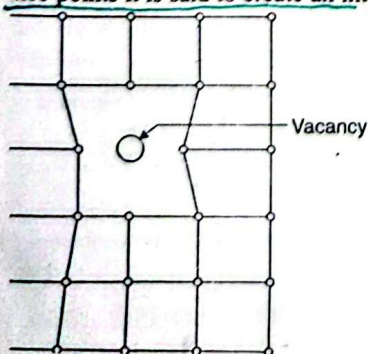


Fig. 13.29

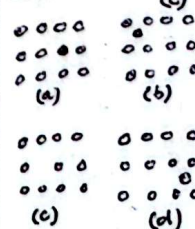
13.16.1 Vacancies

Vacancies are created during crystallisation or from thermal vibrations of the atoms at high temperatures. During thermal vibration, the atoms may acquire sufficiently high energy and evaporate partially or completely and hence create a vacancy (vacancies) in the lattice (Fig. 13.29).

In an ionic crystal, the formation of vacancy requires the charge neutrality which should be maintained in the crystal as a whole. As a

plane defects
① stacking fault
② Grain boundaries

Point defects
are of three basic kinds: the substitutional (a) the interstitial (d) and the vacancy (c).



The defect is due to the presence of foreign atom in an interstitial position.

a turn table. A movable arm is also attached to the turn table for detecting the reflected X-ray beam from the crystal C. This turn table is capable of rotation about a vertical axis and the angle of rotation can be measured on the circular scale. The rates of rotation of the turn table and the detector (ionisation chamber) arm are such that the ionisation chamber always receives the diffracted beam. When the turn table rotates through an angle θ , the ionisation chamber arm automatically rotates through an angle 2θ with the direction of incident ray. This way the measurements of different diffracted X-ray beams' intensities and angles are recorded. Then using Bragg's condition, we get the interplanar spacing and the structure of the crystal.

13.15.3 Laue Method

The Laue method is useful for the determination of crystal structure. In this method, a single crystal is held stationary in the path of an incident X-ray beam, as shown in Fig. 13.25.

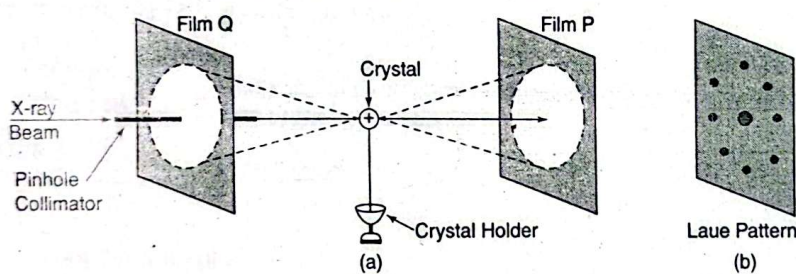


Fig. 13.25

When a continuous X-ray beam through a pinhole is allowed to fall on the crystal, then this beam is diffracted by the crystal, the transmitted - diffracted and the reflected- diffracted beams are received by the films P and Q, respectively as shown in Fig. 13.25a.

The transmitted-diffracted beams form a series of spots, which is the characteristic of crystal structure and is called Laue pattern (Fig. 13.25b). Each spot in the Laue pattern corresponds to the interference maxima for a set of crystal planes satisfying the Bragg's condition ($2d \sin \theta = n\lambda$) for a particular wavelength selected from the beam of incident light. By studying the position and intensities of these Laue spots, the crystal structure can be determined.

13.15.4 Powder Method

It is a standard and straight forward technique for analysing the crystal structure. In this technique, we use the crystal in powder form instead of single crystal so that its tiny crystals (i.e. crystallites) are randomly (i.e. almost continuously) oriented and make all possible angles with the incident beam. A small specimen of the crystalline powder is taken in a small capillary tube (P) of nondiffracting material and is placed in the path of fine monochromatic beam of X-ray (Fig. 13.26). Thus all

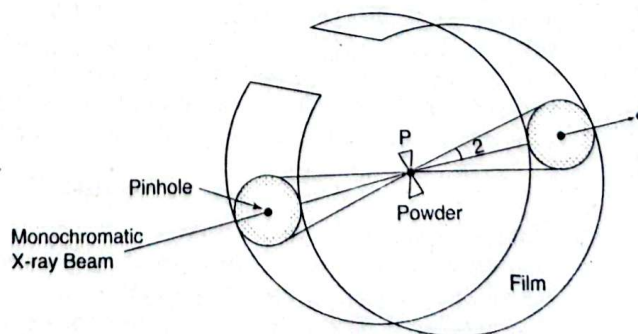


Fig. 13.26

X-rays were discovered by Roentgen in 1895. X-rays are electromagnetic waves of short wave lengths in the range of $10\text{ \AA}$ to $0.5\text{ \AA}$. X-rays are produced when fast moving electrons are suddenly stopped by a solid target. W. H. Bragg and W. L. Bragg studied X-ray diffraction in detail. When monochromatic X-rays impinge upon the atoms in a crystal lattice, each atom acts as a source of scattering radiation of the same wavelength. The crystal acts as a series of parallel reflecting planes. The intensity of the reflected beam at certain angles will be maximum when the path difference between two reflected waves from two different planes is an integral multiple of λ .

16

13.15 CRYSTAL STRUCTURE ANALYSIS

As we know that the X-rays can penetrate solids. Since the wavelength of the X-rays ($\sim 1\text{ \AA}$) is of the order of interplanar spacing, these rays get strongly diffracted from different crystal planes. By analysing the different X-rays, we can gather information about the internal atomic arrangement of a crystal.

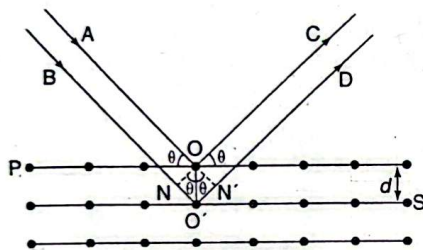


Fig. 13.23

13.15.1 Bragg's Law: Diffraction of X-rays

The diffraction of X-rays from a single crystal is shown in Fig. 13.23. When a narrow beam of X-rays of wavelength λ is incident on the parallel planes at an angle θ , the X-rays are scattered by the atoms of the crystals. Consider a set of parallel lattice planes which are separated by a distance d . Further, two X-ray beams AO and BO' are incident on the crystal, as shown in the figure. The incident X-rays get diffracted at an angle θ at the point O and O' of the crystal planes. The path difference between these two beams can be obtained by drawing perpendiculars ON on BO' and ON' on O'D, as

$$NO' + O'N' = d \sin \theta + d \sin \theta = 2d \sin \theta$$

For constructive interference (or for maximum intensity), we must have

$$2d \sin \theta = n\lambda \quad (i)$$

where n is an integer. The above relation is known as Bragg's law of diffraction. It is useful in calculating the distance d between crystal lattice planes once we know the wavelength λ of the X-rays and measure the angle of diffraction θ .

It is clear from Bragg's condition (i) that every X-ray will not get diffracted by the atoms of a crystal, only those will be diffracted whose wavelength λ and the angle θ exactly match this condition. The standard methods of X-ray diffraction used in the analysis of crystal structure are designed to achieve this. Bragg's X-ray spectrometer, Laue method and Powder method are such methods, which are discussed below.

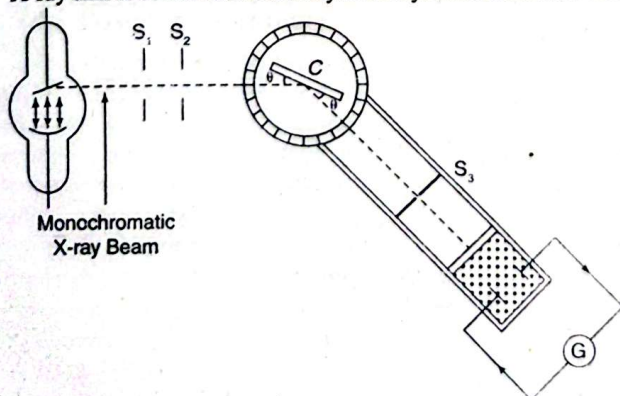


Fig. 13.24

13.15.2 Bragg's X-ray Spectrometer

Bragg devised an apparatus used to study the glancing angle θ and the intensities of diffracted X-rays for a given sample. The essential parts of Bragg's spectrometer are shown in Fig. 13.24.

A monochromatic X-ray beam from the source is passed through two slits, S_1 and S_2 which collimate it into a fine narrow beam. This X-ray beam is then allowed to fall on the crystal sample C mounted at the centre of

However, it can be stronger for those metals in which the number of valence electrons is greater. They are good conductors. Because of the presence of free electrons, they have high thermal and electrical conductivities. Most of the atoms in the first four groups of the periodic table like Li, Na, Cu, Ag, Zn, Fe, etc. are good examples in which metallic bond exists.

13.14.4 Molecular Bond or van der Waals Bond

Many solids or crystals are composed of neutral atoms or molecules without any transferring or sharing of the electrons. Hence, the ionic, covalent and metallic bonds are not possible between these atoms. These substances remain bound by much

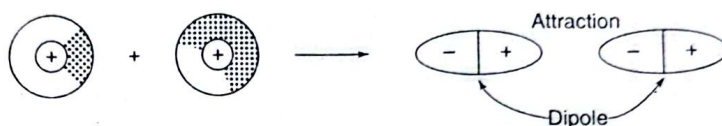


Fig. 13.21

weaker short-range attractive forces, which are called van der Waals forces. These forces are weaker than the atomic bonding forces. The van der Waals bonds are usually found in inert gases in which outermost electron orbits are completely filled, i.e., there are no valence electrons and hence they are incapable of forming any bond. These bonds are formed due to electrostatic attraction between oscillating or permanent dipoles (Fig. 13.21). As we know that the dipoles are formed due to the asymmetrical charge distribution around atoms, these dipoles are called oscillating dipoles.

In the case of inert gases, there is a very small attraction between the atoms due to closed outer shells. These gases condense when the temperature is reduced and hence a weak interatomic attraction is developed due to van der Waals forces. The van der Waals forces are non-directional in nature and a little energy is required to break the bonds because these are much weaker than ionic and covalent bonds. These types of bonds are found in inert gases like solid argon and in many organic symmetrical molecules like methane (CH_4).

13.14.5 Hydrogen Bond

In certain crystals, a positive hydrogen ion (H^+) attracts negative ions such as F^- , O^- , N^- , etc. Sometimes, due to electrostatic attraction, attachment between atoms in different molecules or within a molecule occurs in addition to bonds, which holds atoms together to form molecules. A hydrogen bond is formed when a hydrogen atom makes such an attachment or association with an electronegative atom

like oxygen, nitrogen, fluorine, etc. The hydrogen bond is found in H_2O , HF and in many organic molecules, particularly proteins and DNA molecules. In the water molecule (H_2O), the hydrogen and oxygen atoms are held together by covalent bonds. The positive dipole end i.e. hydrogen, can strongly attract the negative dipole end of water molecule. This bonding of the water molecule is shown in Fig. 13.22, where dashed lines represent the hydrogen bond.

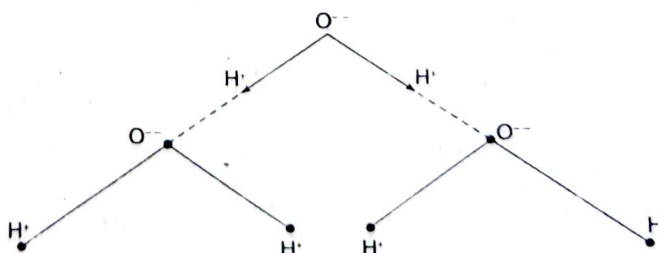


Fig. 13.22

13.14 Engineering Physics

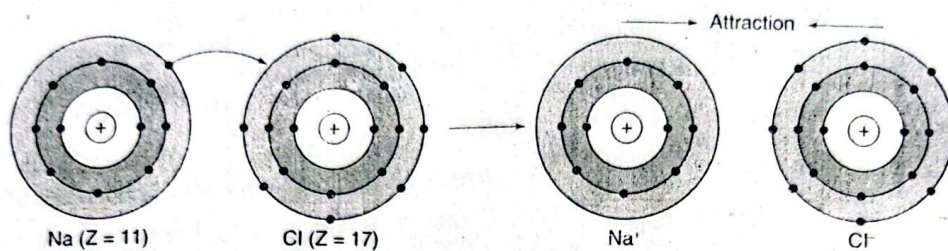


Fig. 13.19

The ionic bonds are strong, hard and brittle. The cohesive energy of ionic crystals is very high. Thus the ionic crystals have very high melting point and possess high latent heat of fusion. The ionic crystals are insulators in general because of their very low conductivity at ordinary temperature. The conductivity of ionic crystals increase with increase in temperature. Many ionic crystals are soluble in water (polar liquid) but not soluble in nonpolar liquids like ether.

13.14.2 Covalent Bond

In covalent crystals, the valency electrons do not get transferred from one atom to another as happens in an ionic bond but are shared equally by both of the atoms. This sharing of valence electrons of the constituent atoms forms a covalent bond. The sharing of electrons takes place in such a way that an electron with spin up pairs with an electron with spin down, if that electron can occupy the states as per Pauli's exclusion principle. The sharing of one pair of electrons forms a singlet covalent bond and a double bond is obtained when two pairs of electrons are shared. The covalent bond is also known as a homopolar or electron pair bond. The conductivity of a covalent bond is low and increases with temperature. These are directional bonds and are very hard because the bond is very strong, for example in diamond.

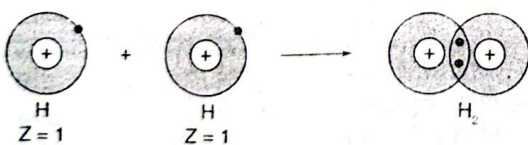


Fig. 13.20

The simplest example of covalent bonding is the H_2 molecule, as shown in Fig. 13.20. As the hydrogen atoms come close to each other, each of the two electrons are attracted by both the nuclei. In case of an oxygen molecule, two oxygen atoms share two pairs of electrons, thus forming a covalent double bond. A triple bond is formed by the sharing of three pairs of electrons in a nitrogen molecule. Covalent bonds are also formed between atoms of different elements like HCl , H_2O , NH_3 , etc.

13.14.3 Metallic Bond

Metallic crystals are commonly known as metals. In the atoms of metals, the electrons in the outermost orbits are loosely bound as the ionisation energy is low in the case of metals. These electrons are free to move around among all the atoms and are called free electrons or conduction electrons. This way the metals have residual positive ions. The electrostatic attraction between these positive ions and negative electron gas is responsible for holding the solid together. This type of bonding is called metallic bonding.

The metallic bond is electrostatic in nature, though partially, and do not exert directional influence. The metallic bond is weaker in nature than the covalent bond because of the fewer electrons bonding the nuclei.

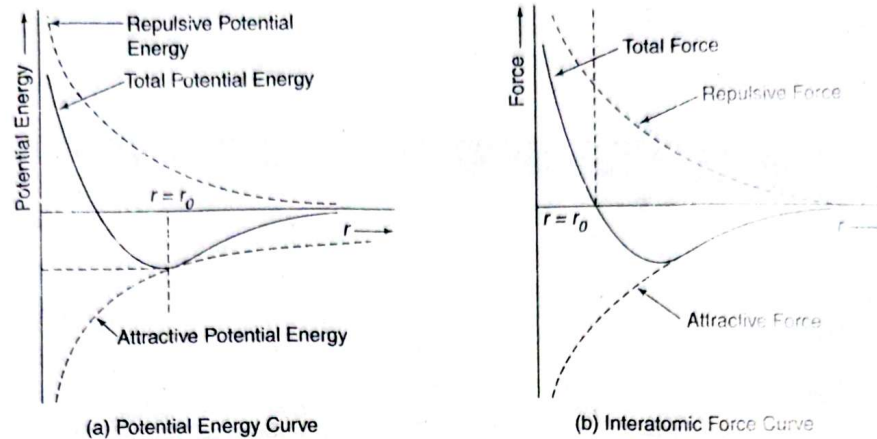


Fig. 13.18

For the fixed value of α , β , n and m , it is clear from this equation that the force $F = 0$ at a particular distance $r = r_0$. The potential energy and interatomic force curves are shown in Fig. 13.18.

When the distance r between the two atoms is very large, then it is clear from the above figures that no force acts between the atoms and hence the total potential energy is also zero. If the atoms approach each other and come close together nearly equal to the atomic diameter then repulsive force also begins to act and attractive force also gets very large. If the atoms come even more closer to each other then the repulsive force increases faster and the net force between atoms becomes repulsive in nature.

At a particular distance ($r = r_0$), the attractive and repulsive forces attain equal values and therefore no net force acts between the atoms. This position is known as equilibrium position in which the molecule becomes most stable because of the minimum potential energy.

13.14 DIFFERENT TYPES OF BONDING FORCES

The constituent particles of crystals have different types of charge distribution which provides different types of binding forces. The binding forces in most cases are electrostatic in nature but the distribution of electrons in various atoms are qualitatively different in different crystals. These binding forces are of different types, for example, ionic bond, covalent bond, metallic bond, molecular bond (or van der Waals bonds) and hydrogen bonds.

13.14.1 Ionic Bond

The ionic bond is formed due to transfer of one or more electrons from one type of atoms that lose electrons readily to the other type that have affinity for electrons. Due to the transfer of electrons, these atoms become positive and negative ions. The two types of atoms, which are involved in ionic bonding, are of different types. The arrangement of ions formed in the ionic bonding is such that the Coulomb attraction between ions of opposite charges is stronger than the Coulomb repulsion between ions of the same charges. Thus, the ionic bond results from the electrostatic interaction of oppositely charged ions. In this situation, both the ions attract each other and form ionic bond, which is clearly shown in Fig. 13.19. The examples of ionic crystals are NaCl, CsCl, KBr, KOH, etc. The ionic bond is nondirectional in nature.

Therefore, atomic packing fraction $f = \frac{6 \times \frac{4}{3} \pi \left(\frac{a}{2}\right)^3}{3\sqrt{2}a^3} = \frac{\pi}{3\sqrt{2}} = 0.74$

or the packing fraction is 74%.

13.13 POTENTIAL ENERGY CURVE AND NATURE OF INTERATOMIC FORCES

The interatomic forces between atoms of the solids are electrostatic in nature which can either be attractive or repulsive.

(a) Interatomic Attractive Forces

These forces act between the atoms to hold them together. These forces are long range and act between the atoms at a finite distance. These forces become zero for infinite interatomic distances.

(b) Interatomic Repulsive Forces

A repulsive force also acts between the atoms to prevent the merging with each other and keep them at a finite distance. These forces are short -range and act only when the atoms come so close with each other that their electron cloud overlap with one another.

To understand interatomic forces, we consider a pair of atoms A and B, which are separated by a distance r at any time. Then, the potential energy due to force of attraction is given by

$$U_{\text{attr}} = -\frac{\alpha}{r^n} \quad (i)$$

and the potential energy due to force of repulsion is given by

$$U_{\text{rep}} = +\frac{\beta}{r^m} \quad (ii)$$

The total potential energy of the system is therefore given by

$$U = U_{\text{attr}} + U_{\text{rep}} \quad (iii)$$

$$U = -\frac{\alpha}{r^n} + \frac{\beta}{r^m}$$

where α , β , n and m are the characteristic constants of the molecule. From the above equation it is clear that if $r = \infty$, then $U = 0$.

The force acting between these two atoms can be written as

$$F = -\frac{dU}{dr} = -\frac{d}{dr} \left[-\frac{\alpha}{r^n} + \frac{\beta}{r^m} \right]$$

$$\text{or } F = -\frac{n\alpha}{r^{n+1}} + \frac{m\beta}{r^{m+1}} \quad (iv)$$

13.12.3 Face Centered Cubic (fcc) Structure

In a face centered cubic lattice, the number of atoms per unit cell = 4

The atomic radius, $r = \frac{\sqrt{2}}{4} a$

Volume of atoms occupying the unit cell = $4 \times \frac{4}{3} \pi r^3 = 4 \times \frac{4}{3} \pi \left(\frac{\sqrt{2}}{4} a \right)^3$

Volume of the unit cell = a^3

Therefore, atomic packing fraction $f = \frac{4 \times \frac{4}{3} \pi \left(\frac{\sqrt{2}}{4} a \right)^3}{a^3} = \frac{\pi}{3\sqrt{2}} = 0.74$.

So, the atomic packing fraction is 74%.

13.12.4 Diamond Structure

In the diamond structure, the number of atoms per unit cell = 8

The atomic radius, $r = \frac{\sqrt{3}}{8} a$

Volume of atoms occupying the unit cell = $8 \times \frac{4}{3} \pi r^3 = 8 \times \frac{4}{3} \pi \left(\frac{\sqrt{3}}{8} a \right)^3$

Volume of the unit cell = a^3

Therefore, atomic packing fraction $f = \frac{8 \times \frac{4}{3} \pi \left(\frac{\sqrt{3}}{8} a \right)^3}{a^3} = \frac{\sqrt{3}}{16} \pi = 0.34$

or the packing fraction is 34%.

13.12.5 Hexagonal Closed Packed (hcp) Structure

In the hexagonal cell, the number of atoms per unit cell = 6

The atomic radius, $r = \frac{a}{2}$

Volume of atoms occupying the unit cell = $6 \times \frac{4}{3} \pi r^3 = 6 \times \frac{4}{3} \pi \left(\frac{a}{2} \right)^3$

Volume of the hexagonal unit cell = $\frac{3\sqrt{3}}{2} a^2 c$ where $c = a\sqrt{\frac{8}{3}}$

$$\begin{aligned} \therefore \text{Volume} &= \frac{3\sqrt{3}}{2} a^2 \sqrt{\frac{8}{3}} a \\ &= 3\sqrt{2} a^3 \end{aligned}$$

13.10 Engineering Physics

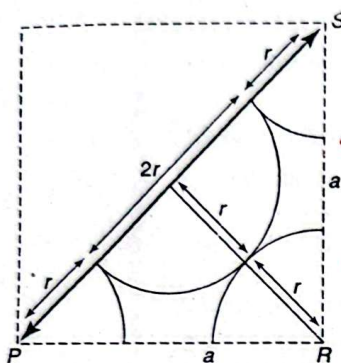


Fig. 13.17

For the face centered cubic structure (fcc), shown in Fig 13.17, one can easily obtain

$$\begin{aligned} PS^2 &= PR^2 + RS^2 = a^2 + a^2 = 2a^2 \\ \therefore PS &= \sqrt{2}a \\ \therefore 4r &= \sqrt{2}a \\ \therefore r &= \frac{a}{2\sqrt{2}} \end{aligned}$$

13.12 PACKING FRACTION

Atomic packing fraction is defined as the ratio of volume of atoms occupying the unit cell to the volume of the unit cell relating to that substance. It is also known as relative packing density. It is denoted by f . We can calculate f for different cell structures.

13.12.1 Simple Cubic (sc) Structure

In a simple cubic cell structure, the number of atoms in unit cell = 1.

The atomic radius is given by half of the lattice constant, i.e. $r = a/2$.

$$\text{Volume of atom occupying the unit cell} = 1 \times \frac{4}{3} \pi r^3 = \frac{4}{3} \pi \left(\frac{a}{2}\right)^3$$

$$\text{Volume of the unit cell} = a^3$$

$$\text{Therefore, atomic packing fraction } f = \frac{\frac{4}{3} \pi \left(\frac{a}{2}\right)^3}{a^3} = \frac{\pi}{6} = 0.52.$$

It also means that the packing fraction is 52%.

13.12.2 Body Centered Cubic (bcc) Structure

In a body centered cubic lattice, the number of atoms per unit cell = 2

$$\text{The atomic radius, } r = \frac{\sqrt{3}}{4} a$$

$$\text{Volume of atoms occupying the unit cell} = 2 \times \frac{4}{3} \pi r^3 = 2 \times \frac{4}{3} \pi \left(\frac{\sqrt{3}}{4} a\right)^3$$

$$\text{Volume of the unit cell} = a^3$$

Therefore, the atomic packing fraction

$$f = \frac{2 \times \frac{4}{3} \pi \left(\frac{\sqrt{3}}{4} a\right)^3}{a^3} = \frac{\sqrt{3}}{8} \pi = 0.68$$

or it is 68%.

we get

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$d^2 \left[\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2} \right] = 1$$

$$\text{or } d = \frac{1}{\sqrt{\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}}$$

For cubic crystal $a = b = c$, we get

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

For tetragonal crystal $a = b \neq c$, we get

$$d = \frac{1}{\sqrt{\frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2}}}$$

13.11 NEAREST NEIGHBOUR DISTANCE AND ATOMIC RADIUS

The distance between the centers of two neighbouring atoms is called nearest neighbour distance. For a closely packed crystal, this distance would be $2r$ for an atom of radius r . The distance r is called atomic radius, which can be represented in terms of edge of cube a for certain unit cell structures. For example, for simple cubic (sc) structure the diameter of an atom would be equal to the cube edge. That is $2r = a$ or $r = a/2$. However, in case of a body centered cubic (bcc) structure, shown in Fig. 13.16, the atomic arrangement is such that

$$PQ = r + 2r + r$$

$$\text{Since } PR = RS = a, \quad PS = a\sqrt{2}$$

$$\text{Therefore, } PQ = \sqrt{PS^2 + SQ^2}$$

$$\text{or } (4r)^2 = 2a^2 + a^2$$

$$\text{or } r = \frac{a\sqrt{3}}{4}$$

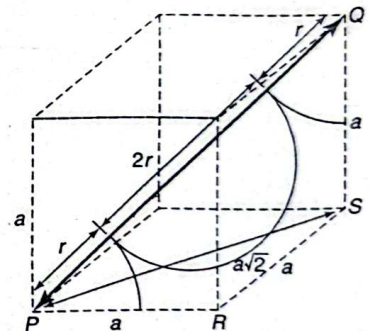


Fig. 13.16

Atomic radius : It is defined as half the distance between two nearest neighbouring atoms in the crystal of pure element.

$$\text{Number of atoms per unit cell, } N = N_i + \frac{N_f}{2} + \frac{N_p}{8}$$

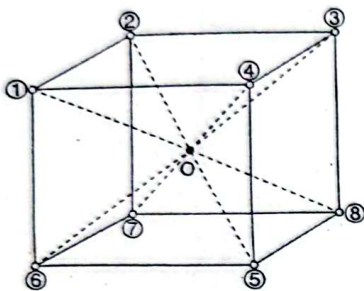


Fig. 13.13

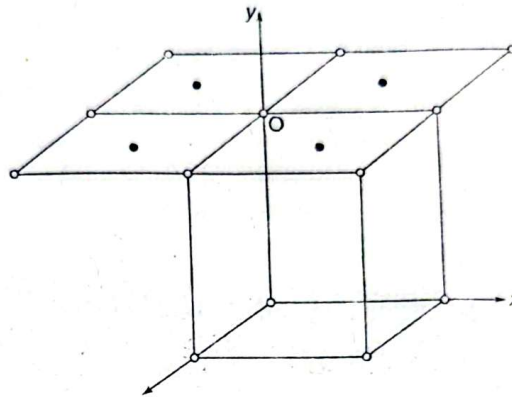


Fig. 13.14

13.9.3 Face Centered Cubic Lattice

It contains lattice points at the centre of each face as well as at all eight corners. Any corner atom has 4 face centered atoms of the surrounding in its own plane and 4 face centered atoms above and below this plane (Fig. 13.14). Hence the coordination number for this case is $12 (= 4 \times 3)$.

13.10 INTERPLANAR SPACING

Let us consider a set planes ($h k l$) in a unit cell. This plane PQR makes intercepts $a/h, b/k, c/l$ on the three Cartesian co-ordinate axes P, Q and R respectively, as shown in Fig. 13.15. Consider O as the origin at a lattice point.

Let the plane PQR ($h k l$) be parallel to the plane passing through the origin. Draw a perpendicular ON from O (origin) to the plane. Thus $ON = d$, is the distance between adjacent planes or interplanar spacing between parallel planes. The normal ON makes angles α, β and γ with the x, y and z axes, respectively. From Fig. 13.15 we obtain $OP = a/h$, $OQ = b/k$ and $OR = c/l$.

$$\text{Now from } \triangle ONP, \frac{ON}{OP} = \cos \alpha = \frac{d}{a/h} = \frac{dh}{a}$$

$$\text{From } \triangle ONQ, \frac{ON}{OQ} = \cos \beta = \frac{d}{b/k} = \frac{dk}{b}$$

$$\text{From } \triangle ONR, \frac{ON}{OR} = \cos \gamma = \frac{d}{c/l} = \frac{dl}{c}$$

Then according to the law of direction cosines,

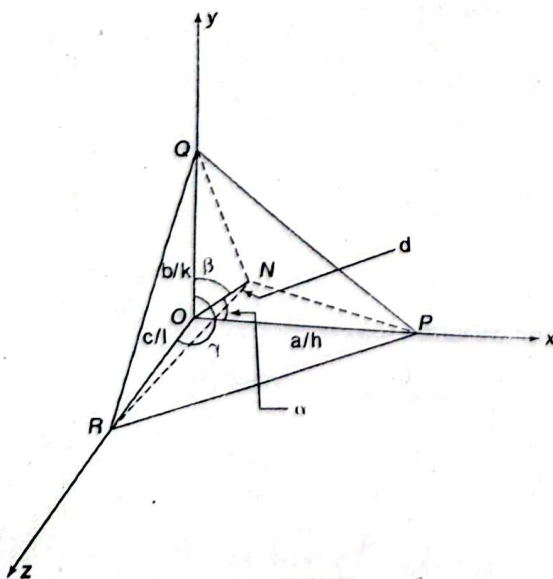


Fig. 13.15

The coordinates of Cl ions

$$\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right), \left(0, 0, \frac{1}{2}\right), \left(0, \frac{1}{2}, 0\right), \left(\frac{1}{2}, 0, 0\right)$$

13.8.2 Cesium Chloride (CsCl) Structure

In a CsCl crystal, we have a body centered cubic lattice, in which the Cs atom is placed at the origin (0, 0, 0), i.e. at the body center point and Cl atoms are placed at the corners of the bcc lattice whose coordinates are given by $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$. Structure of cesium chloride is shown in Fig. 13.10.

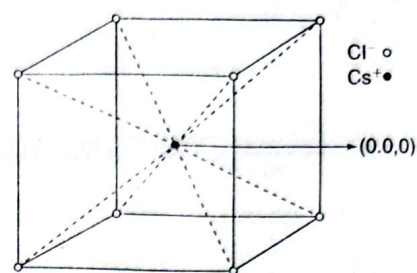


Fig. 13.10

13.8.3 Diamond Structure

The space lattice of diamond is fcc. In diamond structure we have two fcc lattices placed at (0, 0, 0) and $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$ which superimpose each other. In diamond structure we have two carbon atoms placed at (0, 0, 0) and $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$. The diamond crystal structure is shown in Fig. 13.11.

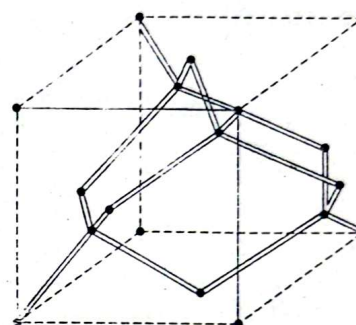


Fig. 13.11

13.9 COORDINATION NUMBER

In a crystal, every atom is surrounded by the other atoms. The number of nearest neighbours to the given atom in crystal lattice is known as coordination number.

13.9.1 Simple Cubic Lattice

This simple cubic lattice contains lattice points at all eight corners of the unit cell. Any corner atom has four neighbours. In the same plane as well as other two neighbours. One of them is exactly above and the other exactly below in the vertical plane (Fig. 13.12). Hence, the coordination number in a simple cubic lattice is 6 (= 4+2).

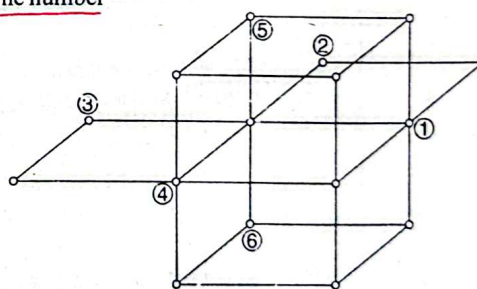


Fig. 13.12

13.9.2 Body Centered Cubic Lattice

It contains one additional lattice point at the center of the body of the unit cell in addition to those at eight corners. In this lattice, every centered atom is surrounded by eight equidistant neighbours (Fig. 13.13). Hence, coordination number in a body centered cubic lattice is 8.

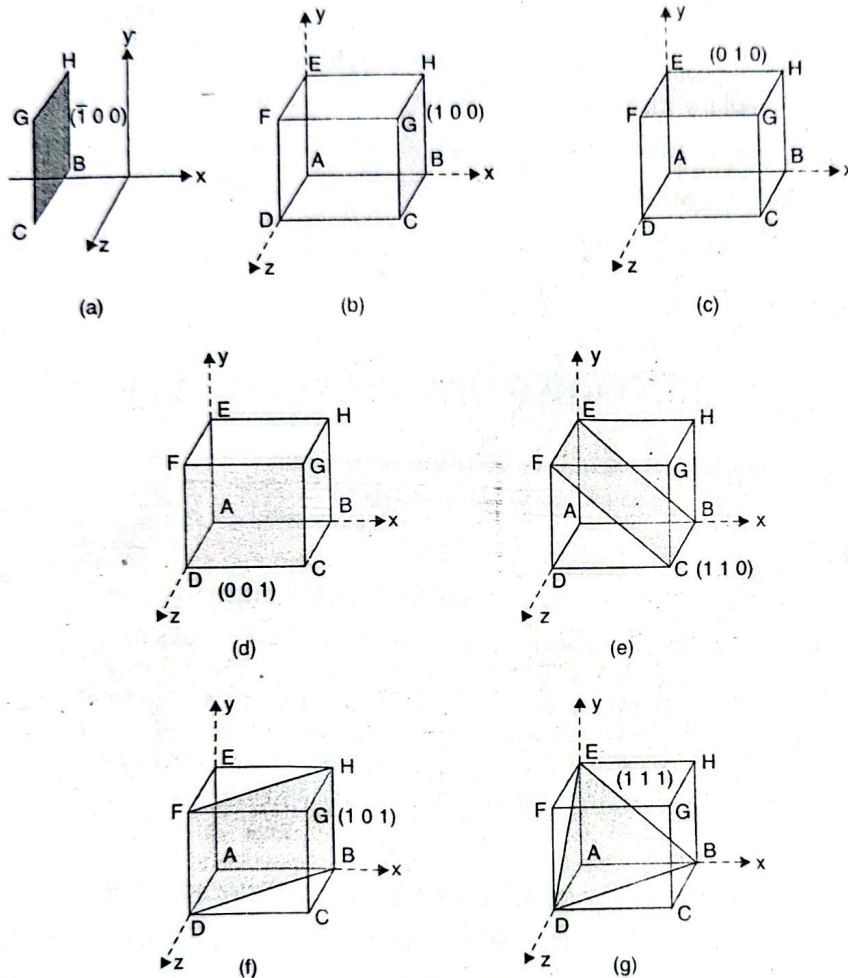


Fig. 13.8

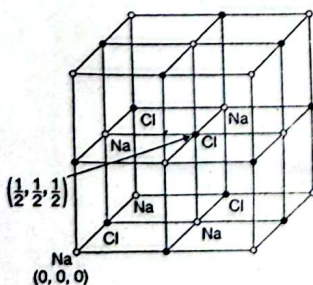


Fig. 13.9

of the cube say at the point $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$. The space lattice is therefore truly fcc, with a basis of one Na ion and one Cl ion separated by one half the body diagonal of a unit cube. There are four Na-Cl ion pairs in each unit cube, with different ions in the positions.

The coordinates of Na ions are defined as

$$(0, 0, 0), \left(\frac{1}{2}, \frac{1}{2}, 0\right), \left(\frac{1}{2}, 0, \frac{1}{2}\right), \left(0, \frac{1}{2}, \frac{1}{2}\right)$$

The coordinates of

13.8.2 Cesium

In a CsCl crystal, the Cs atom is placed at the center point and Cl atoms at the corners.

whose coordination number is 8.

13.8.3 Diamond

The space lattice of diamond is two fcc lattices displaced by one half the body diagonal of the cube. In diamond, the atoms are located at $(0, 0, 0)$ and $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$.

Fig. 13.11.

13.9 Crystallography

In a crystal, every atom has a number of nearest neighbours. This number is known as coordination number.

13.9.1 Simple Cubic

This simple cubic lattice has one atom at each of the eight corners of the cube. Each atom has six nearest neighbours. One is exactly below and one is exactly above the atom. The coordination number is 6 ($= 4 + 2$).

13.9.2 Body-Centered Cubic

It contains one atom at each of the eight corners and one atom at the center of the cube. In this lattice, the coordination number is 8.

Let us assume that a , b and c are the magnitudes of fundamental translation vectors along the three axes, respectively. Again we consider that ABC represents the plane whose Miller indices are to be obtained. Let OA , OB and OC are the intercepts made by this plane along the three axes (Fig. 13.7).

In order to find the Miller indices, we do the following.

- We find the intercepts along the three axes.
- We express these intercepts as multiple of lattice parameters.
- We divide these intercepts by lattice parameters.
- We take the reciprocal of these.
- We clear these fractions by taking the LCM of the denominators and multiplying it. This gives the required Miller indices.

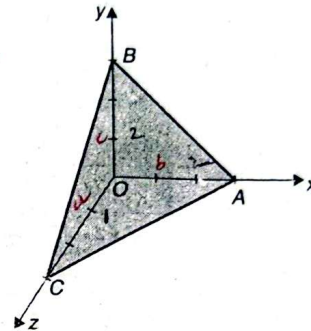
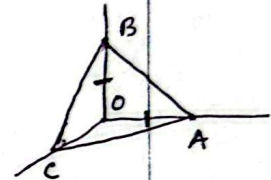


Fig. 13.7



This can be understood in a better way in the following steps corresponding to the above steps a to e.

Directions	x	y	z
Step (a)	OA	OB	OC
Step (b)	pa	qb	rc
Step (c)	$\frac{pa}{a}$	$\frac{qb}{b}$	$\frac{rc}{c}$ or $p \quad q \quad r$
Step (d)	$\frac{1}{p}$	$\frac{1}{q}$	$\frac{1}{r}$
Step (e)	(h)	(k)	(l)

orientation $(2 \ 2 \ 1) \approx \frac{1}{2} : \frac{1}{2} : 1 \approx (1 \ 1 \ 2) \rightarrow$ Miller indices

$(4 \ 4 \ 2) \approx \frac{1}{4} : \frac{1}{4} : \frac{1}{2} \approx (1 \ 1 \ 2)$

$(\frac{1}{2}, \frac{1}{2}, \frac{3}{4}) \approx (\frac{1}{1/2} : \frac{1}{1/2} : \frac{1}{3/4})$

$\approx (2 : 2 : \frac{4}{3}) \approx (6 \ 6 \ 4)$

LCM: Least common multiple $\approx (3 \ 3 \ 2)$

3.7.1 Significance of Miller Indices

Same Miller indices are assigned to all parallel planes, for example, planes with coordinates x, y, z ; $2x, 2y, 2z$; $-x, -y, -z$ etc., are represented by the same Miller indices. The corresponding index for a plane, which is parallel to one coordinate axis, is zero. The corresponding index is negative for a plane that cuts an axis on the negative side of the origin. The negative index is represented by putting a bar above the index.

3.7.2 Representation of Miller Indices

Let us consider a simple cubic system ($\alpha = \beta = \gamma = 90^\circ$ and $a = b = c$). For this system in Fig. 13.8a, x, c, d, e, f and g , we show Miller planes corresponding to $(\bar{1} \ 0 \ 0)$, $(1 \ 0 \ 0)$, $(0 \ 1 \ 0)$, $(0 \ 0 \ 1)$, $(1 \ 1 \ 0)$, $(1 \ 0 \ 1)$ and $(1 \ 1 \ 1)$ planes in a cubic crystal respectively.

13.8 SIMPLE CRYSTAL STRUCTURES

Below we discuss simple crystal structures, for example, the structures of sodium chloride, cesium chloride and diamond.

13.8.1 Structure of NaCl

The sodium chloride structure is shown in Fig. 13.9. It consists of two face centered cubic sublattices, one of Na ion having its origin at the point $(0, 0, 0)$ and the other of the Cl ion having its origin midway along a edges

13.4 Engineering Physics

13.4.1 Cubic System

In a cubic system, there are three types of lattices, namely simple cubic, body centered cubic and face centered cubic. In addition to these structures, other structures are also depicted in Fig. 13.4.

- Simple Cubic:** It contains lattice points at all eight corners of the unit cell. It is represented by sc.
- Body Centered Cubic:** It contains one additional lattice point at the centre of the body of the unit cell including at all eight corners. It is represented by bcc.
- Face Centered Cubic:** It contains lattice points at the centre of each face as well as at all eight corners. It is represented by fcc.

13.5 TRANSLATION VECTORS

We take any lattice point O as an origin in a plane lattice shown in Fig. 13.5. Any other point in the two-dimensional lattice can be obtained by repeatedly translating the vectors $\vec{a}$ and $\vec{b}$. These vectors are known as basis vectors. Based on these basis vectors, we obtain the plane lattice by their repeated translation. The position vector of any other lattice point, i.e., translation vector, can be represented as

$$\vec{T} = n_1 \vec{a} + n_2 \vec{b}$$

where n_1 and n_2 are the integers which represent the number of lattice points along the directions OA and OB, respectively, and $\vec{a}$ and $\vec{b}$ are the primitives. In case of three-dimensional crystal structures, the arrangement of points can be represented as

$$\vec{T} = n_1 \vec{a} + n_2 \vec{b} + n_3 \vec{c}$$

where $\vec{a}$, $\vec{b}$ and $\vec{c}$ are the primitives along X, Y and Z axes, respectively.

Fig. 13.5

13.6 LATTICE PLANES

A crystal lattice is made of a large number of parallel equidistant planes (Fig. 13.6) known as lattice planes and can be chosen in a number of ways, as shown in Fig. 13.6a, b, c and d.

13.7 MILLER INDICES

The integers which determine the orientation of a crystal plane in relation to the three crystallographic axes are called Miller indices. In order to find the Miller indices, the reciprocals of the intercepts of the plane on the axes in terms of lattice constants are reduced to the smallest integers in ratio. Miller indices are also called crystal indices.

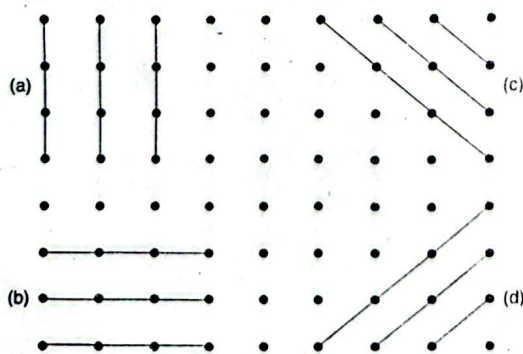


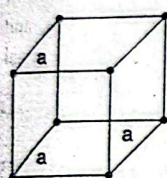
Fig. 13.6

that there are fourteen different types of crystal lattices under the seven crystal systems. These seven types of crystal systems are tabulated in Table 13.1, and shown in Fig. 13.4.

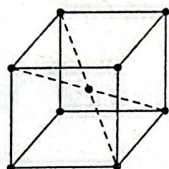
Table 13.1

S.No.	Name of System	Relative Between		Number of Possible Lattices	Examples
		Primitives	Angles		
1.	Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	3(P, F, I)	NaCl
2.	Trigonal	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	1(R)	CaSO ₄
3.	Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	2(P, I)	NiSO ₄
4.	Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$	1(P)	Quartz
5.	Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	4(P, C, F, I)	KNO ₃
6.	Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ, \neq \beta$	2(P, C)	FeSO ₄
7.	Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	1(P)	CuSO ₄

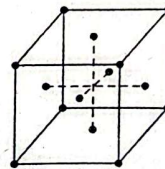
P = Primitive, C = Base centered, I = Body Centered, F = Face centered,
R = Rhombohedral



Simple Cubic



Body Centered Cubic



Face Centered Cubic

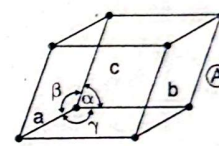
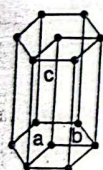
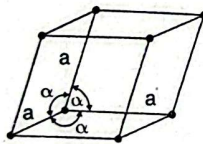
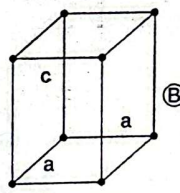
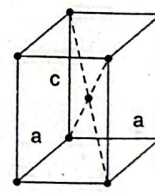
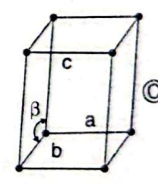
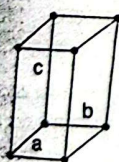
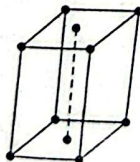
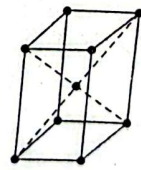
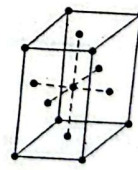
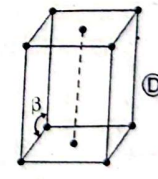
Triclinic
(Primitive)Hexagonal
(Primitive)Trigonal
(Rhombohedral)Tetragonal
(Primitive)Tetragonal
(Body Centered)Monoclinic
(Primitive)Orthorhombic
(Primitive)Orthorhombic
(Base Centered)Orthorhombic
(Body Centered)Orthorhombic
(Face Centered)Monoclinic
(Base Centered)

Fig. 13.4

13.2 Engineering Physics

Even amorphous materials have some short range order at the atomic length scale due to the nature of chemical bonding. They are considered supercooled liquids in which the molecules are arranged in a random manner somewhat as in the liquid state. Glass and plastic are the examples of amorphous solids. Unlike crystalline solids, amorphous solids do not have definite melting points.

13.3 UNIT CELL

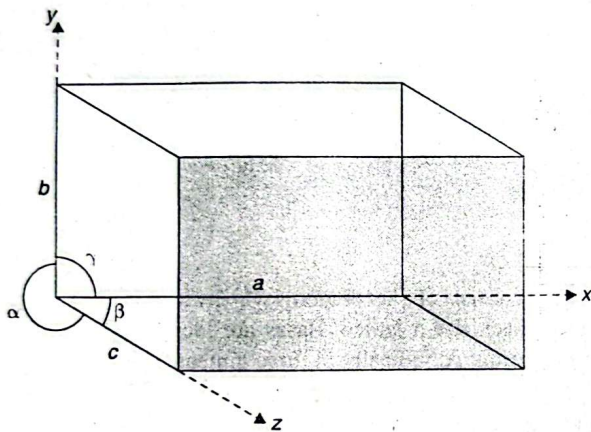


Fig. 13.1

The smallest portion of a space lattice which can generate the complete crystal by repeating its own dimensions in various directions is called a unit cell. A unit cell is defined by the length of its edges and by the angles between them, as shown in Fig. 13.1.

Vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are called lattice vectors that form primitive axes in the crystal structure. We also call them crystallographic axes, as the directions defined by these vectors are nothing but crystal axes. These vectors are used in translation vector and hence are called fundamental translation vectors. The magnitudes of vectors a , b and c are called lattice constants that specify the distances of the bases along the crystal axes.

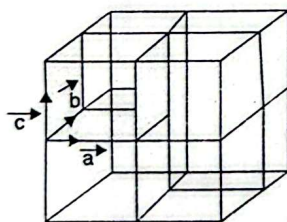


Fig. 13.2

13.3.1 Primitive Lattice Cell

The parallelepiped defined by primitive axes $\vec{a}$, $\vec{b}$, $\vec{c}$ is called a primitive cell. A primitive cell is a type of unit cell which fills all the space under the action of suitable crystal translation operation. A primitive cell is a minimum volume unit cell, as shown in Fig. 13.2. There is a density of one lattice point per primitive cell. The volume of a primitive cell is defined by primitive axes a, b, c as $[(\vec{a} \times \vec{b}) \cdot \vec{c}]$.

13.3.2 Wigner-Seitz Primitive cell

The smallest volume enclosed by the normal lines drawn from midpoints of the lines which connect lattice point to all nearby lattice points is called a Wigner-Seitz primitive cell. It is shown in Fig. 13.3.

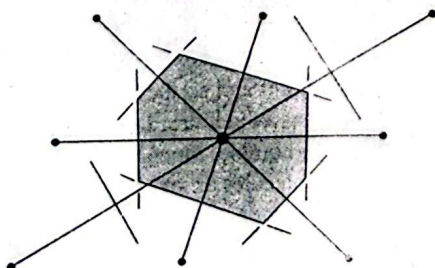


Fig. 13.3

13.4 TYPES OF CRYSTALS

All crystals are classified into seven crystal systems on the basis of the shape of the unit cells. Bravais in 1948 explained

Crystal Structure

13.1 INTRODUCTION

A crystal structure is a unique arrangement of atoms. It consists of a set of atoms which are identical in composition, arrangement and orientation called basis and a lattice. Bases are located upon the points of a lattice, which is an array of points repeating periodically in three dimensions. The points can be thought of as forming identical tiny boxes, called unit cells, that fill the space of the lattice. The lengths of the edges of a unit cell and the angles between them are called the lattice parameters. A crystal structure and symmetry play an important role in determining many of its properties, like electronic band structure and optical properties.

It is clear that a crystal structure is formed by the addition of a basis of atoms to every lattice point. Mathematically, it can be represented as

$$\text{Crystal structure} = \text{Lattice} + \text{Basis}$$

A lattice is a regular periodic arrangement of points in space.

13.2 TYPES OF SOLIDS

Attached to every lattice point there is basis of atoms, with every basis identical in composition, arrangement and orientation.

The solids are broadly classified into two groups, namely, crystalline solids and amorphous solids.

13.2.1 Crystalline Solids

Crystalline solids are arranged in fixed geometric patterns or lattices. Ice, methanol and sodium chloride are a few examples of crystalline solids. They have orderly arranged units and are practically incompressible. Crystalline solids also show a definite melting point and so they pass rather sharply from solid to liquid state. There are various crystalline forms which are divided into seven crystal system or shapes. They are cubic, tetragonal, hexagonal, rhombic, monoclinic, trigonal and triclinic. The units that constitute these systems can be atoms, molecules or ions. Ionic and atomic crystals are hard and breakable with high melting points.

13.2.2 Amorphous Solids

A rigid material whose structure lacks crystalline periodicity is called an amorphous solid. It means the pattern of its constituent atoms or molecules does not repeat periodically in three dimensions.